SHUBHAM KUMAR

B.Tech Computer Science and Engineering (CSE) · Data Science & AI

Delhi, India | +91 73618 57543 | yshubham0103@gmail.com | linkedin.com/in/shubham-kumar-288b7437b | github.com/shubham000111222 | Portfolio

SUMMARY

B.Tech Computer Science and Engineering (CSE) student with a strong foundation in Python, backend development, and AI/ML engineering. Experienced in building end-to-end NLP systems, developing REST APIs with FastAPI, and containerizing applications with Docker. Passionate about Generative AI, LLM applications, prompt engineering, and intelligent automation. Seeking to build practical AI solutions that turn prototypes into reliable, user-focused products.

TECHNICAL SKILLS

Languages & Core: Python, SQL, Git, REST APIs

AI, LLMs & NLP: HuggingFace Transformers, NLP, Prompt Engineering, Generative AI

Backend & Deployment: FastAPI, Docker, PostgreSQL, Streamlit

ML / DL: PyTorch, Scikit-learn, XGBoost, LightGBM

Data & Visualisation: Pandas, NumPy, Matplotlib, Seaborn, Power BI, Plotly

PROJECTS

ResumeRank AI — NLP & Candidate Screening Pipeline
GitHub Live Demo

Python · BERT · TF-IDF · spaCy · Streamlit · Docker

- Engineered an end-to-end NLP pipeline for parsing unstructured documents into structured data and extracting 80+ entities using spaCy NER.
- Evaluated model outputs and continuously refined workflows to ensure high accuracy using Precision@K and NDCG@K metrics.
- Packaged the application using Docker and deployed it for interactive, real-world usage.

Medical Image Classifier API
GitHub Live Demo

PyTorch · FastAPI · Streamlit · REST APIs

- Developed a backend REST API using FastAPI to serve deep learning model predictions for Chest X-Ray diagnosis.
- Fine-tuned EfficientNet-B4 on 112K images, achieving 91.2% accuracy.
- Connected AI to the real world by integrating the FastAPI backend services with an interactive Streamlit frontend.

Customer Churn Prediction & Automation
GitHub Live Demo

XGBoost · FastAPI · PostgreSQL · Docker · Streamlit

- Built and trained an ensemble ML model on a 3M+ row dataset, optimizing for sub-50ms asynchronous inference times.
- Automated data processing workflows to reduce manual effort and improve operational efficiency.
- Served real-time predictions via a robust FastAPI REST endpoint and containerized the entire ecosystem with Docker for scalable deployment.

Financial Sentiment Analyser (NLP)
GitHub Live Demo

FinBERT · HuggingFace · Airflow · PostgreSQL

- Fine-tuned FinBERT (HuggingFace) on 80K financial headlines for accurate sentiment classification.
- Designed and built automated ETL pipelines (Airflow) connecting external web APIs to a PostgreSQL database for continuous data processing.

Retail Demand Forecasting System
GitHub Live Demo

Python · Prophet · LSTM · Keras · Streamlit · Docker

- Built Prophet + LSTM hybrid on Kaggle Store Sales dataset (2.6M rows, 54 stores, 33 product families); reduced MAPE from 22% to 9% vs naive baseline using holiday and promotion regressors.
- Generated 30-day forecasts with 90% confidence intervals across 1,200 SKUs.